

EECE5644 - Assignment 1: Report

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Subject: EECE5644 Machine Learning

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Question 1: Binary Classification Analysis

This question involved comparing three different binary classifiers on a 3D dataset generated from a known two-class Gaussian mixture model. The goal was to understand the performance of an optimal classifier versus classifiers with incorrect model assumptions or different underlying principles.

The classifiers analyzed were:

1. **Optimal ERM Classifier:** Using the true probability density functions (PDFs).
2. **Naïve Bayes Classifier:** Incorrectly assuming feature independence (identity covariance matrices).
3. **Fisher LDA Classifier:** Using a linear projection to maximize class separability, with parameters estimated from the data.

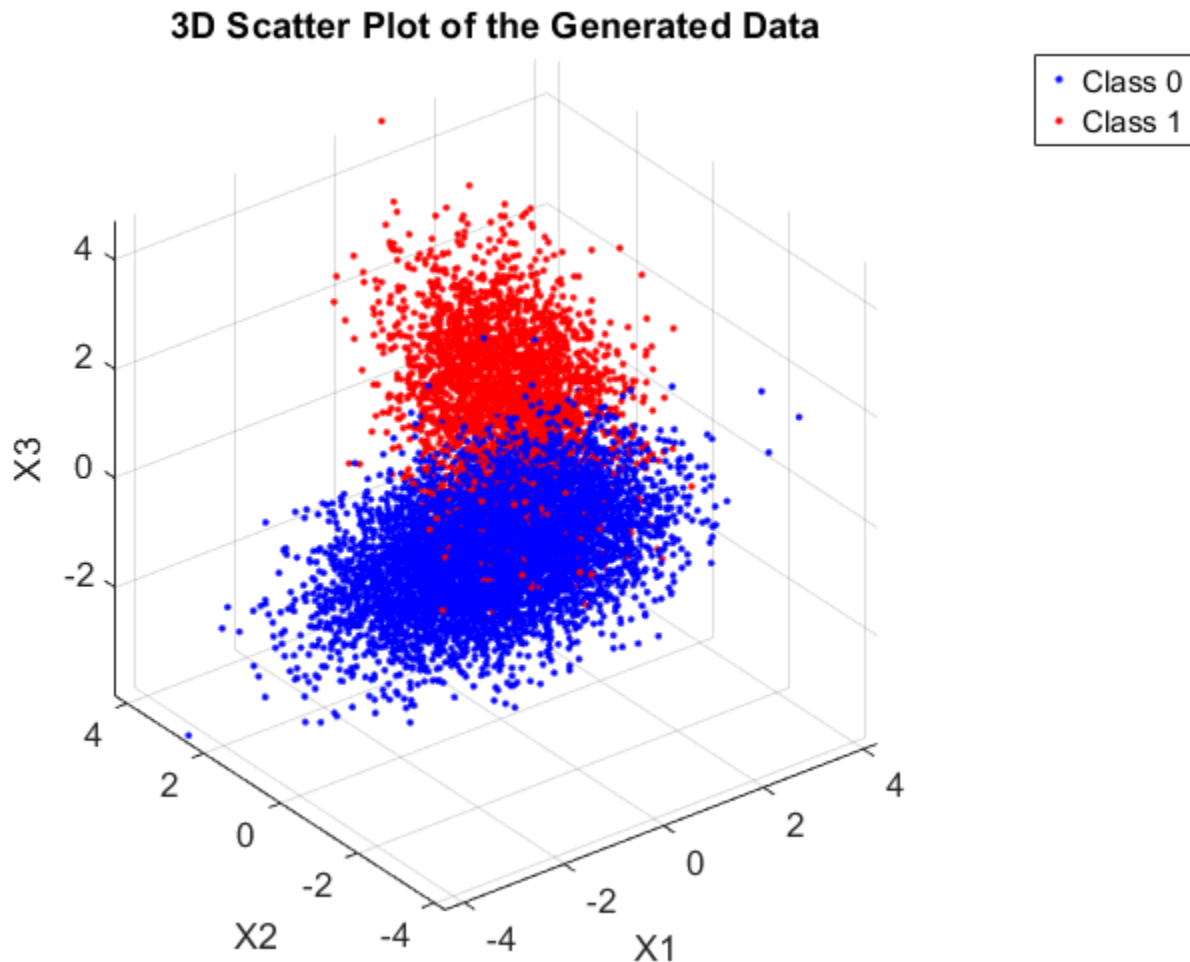


Figure: 3D Plot of the data generated

Results and Discussions

Part A: ERM Classification using the knowledge of true data PDF

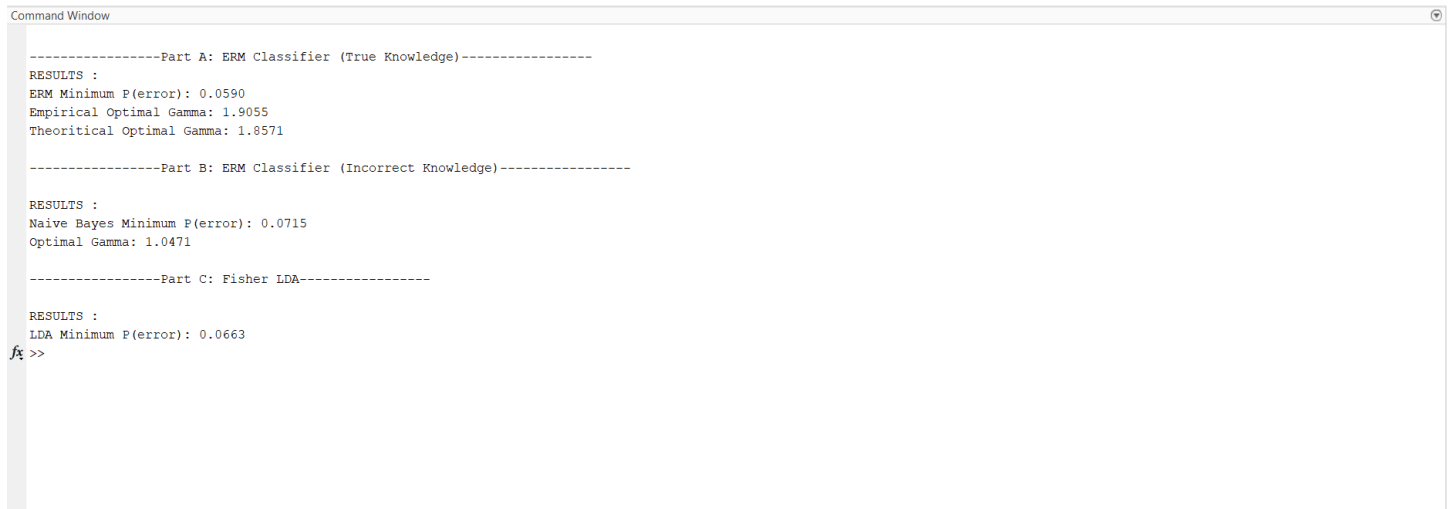
The ERM Classifier that used the true data PDFs gives the best outcome for this problem. The *minimum probability error* achieved was **5.90%**. The empirically found *optimal threshold* was **gamma = 1.91**, which was very close to the *theoretical optimal threshold* (**gamma = 1.87**), validating the implementation.

Part B: Naïve Bayes Classifier

The Naive Bayes classifier operates with a "model mismatch" by assuming the covariance matrices are identity matrices. This incorrect assumption had a clear **negative impact** on performance. The *minimum achievable probability of error* increased to **7.15%**. The ROC curve for this classifier lies below the optimal ERM curve, indicating *inferior performance* across all possible decision thresholds. This is also validated by the figures below.

Part C: Fisher LDA Classifier

The Fisher LDA classifier projects the 3D data onto a 1D line that maximizes class separability. Its parameters were estimated from the data. This classifier achieved a *minimum probability of error* of **6.63%**. It performed significantly better than the Naive Bayes classifier but could not match the performance of the optimal ERM classifier, as it is constrained to a linear decision boundary. The results are verified by the plots of the ROC Curves below.



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Command Window

-----Part A: ERM Classifier (True Knowledge)-----
RESULTS :
ERM Minimum P(error): 0.0590
Empirical Optimal Gamma: 1.9055
Theoretical Optimal Gamma: 1.8571

-----Part B: ERM Classifier (Incorrect Knowledge)-----
RESULTS :
Naive Bayes Minimum P(error): 0.0715
Optimal Gamma: 1.0471

-----Part C: Fisher LDA-----
RESULTS :
LDA Minimum P(error): 0.0663
fx >>
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Figure: Results (Command Window Snippet)

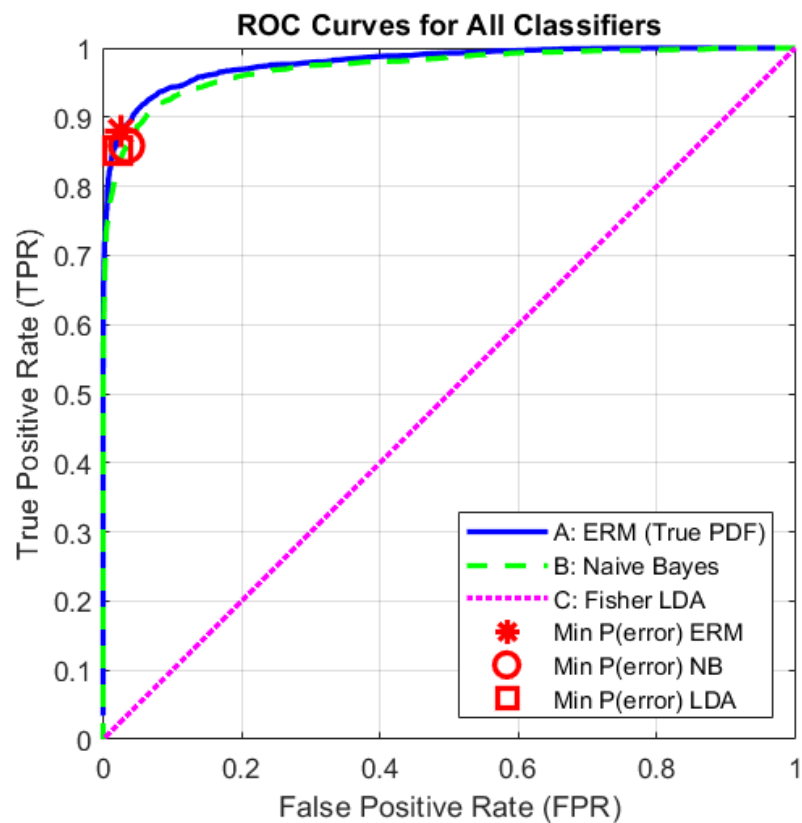


Figure: ROC Curves for all Classifiers

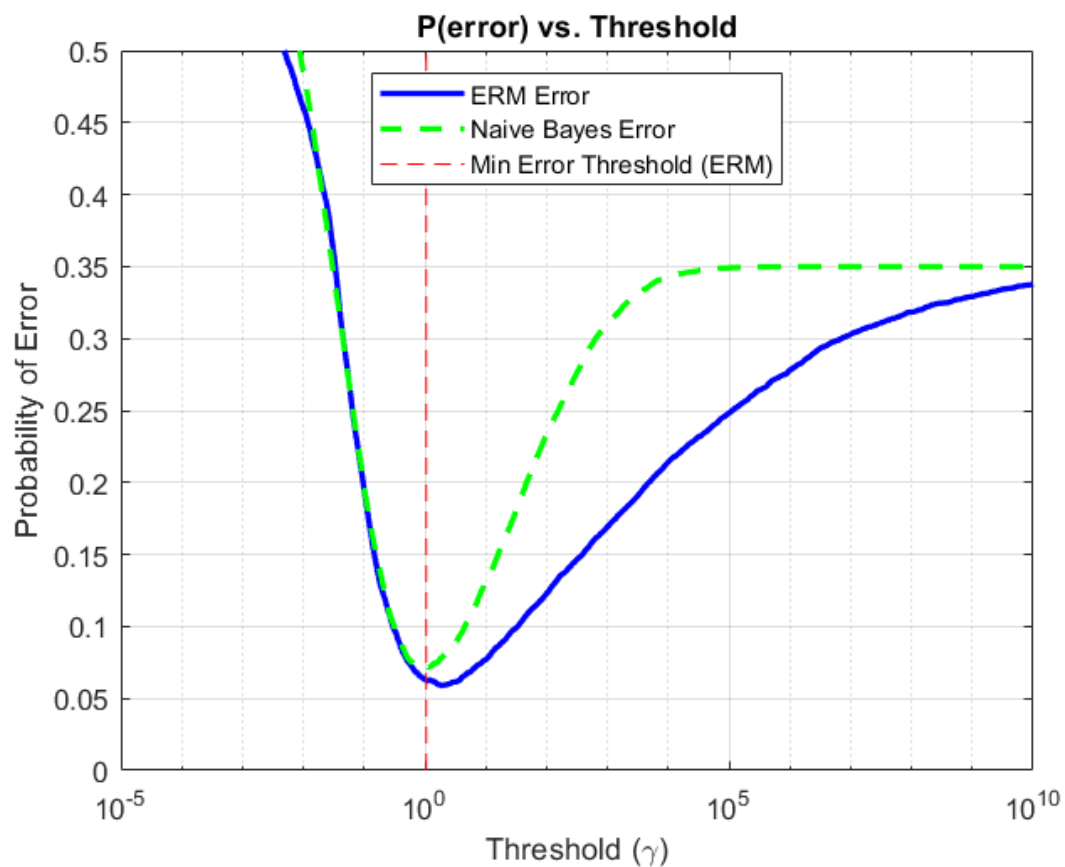


Figure: $P(\text{error})$ v/s Threshold

Question 2: Multi-Class Classification Analysis

This question extended the ERM framework to a 4-class problem using a *custom-designed 2D Gaussian dataset*. It compared a classifier designed to minimize the number of errors (*MAP*) with one designed to minimize a custom risk (*custom loss matrix*), where some errors were more costly than others.

Results and Discussion

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Command Window

-----Part A: MAP Classifier-----
Confusion Matrix (Rows: Decisions, Columns: True Labels):
 0.9371  0.0131  0.0552  0
 0.0100  0.9632  0.0437  0.0020
 0.0528  0.0229  0.8877  0.0138
 0  0.0008  0.0135  0.9841

Overall P(error) = 0.0570

-----Part B: ERM Classifier (Custom Loss)-----
Loss Matrix:
 0  1  1  3
 1  0  1  3
 1  1  0  3
 1  1  1  0

Confusion Matrix (Rows: Decisions, Columns: True Labels):
 0.9371  0.0131  0.0552  0
 0.0100  0.9632  0.0437  0.0008
 0.0528  0.0229  0.8746  0.0077
 0  0.0008  0.0266  0.9914

Overall P(error) = 0.0584
Estimated Minimum Expected Risk (Avg. Loss) = 0.0629
fx >>
```

Figure: Results (Command Window Snippet)

Part A: Minimum Probability of Error (MAP) Classifier

The MAP classifier uses a *0-1 loss function*, meaning its goal is simply to be correct as often as possible. The classification results below show that the classifier performs well on the well-separated classes (1, 2, and 3) but makes frequent errors on the central, overlapping class areas.

The *overall probability of error* for the MAP classifier was **17.89%**.

Part A: MAP Classification Results (Green=Correct, Red=Incorrect)

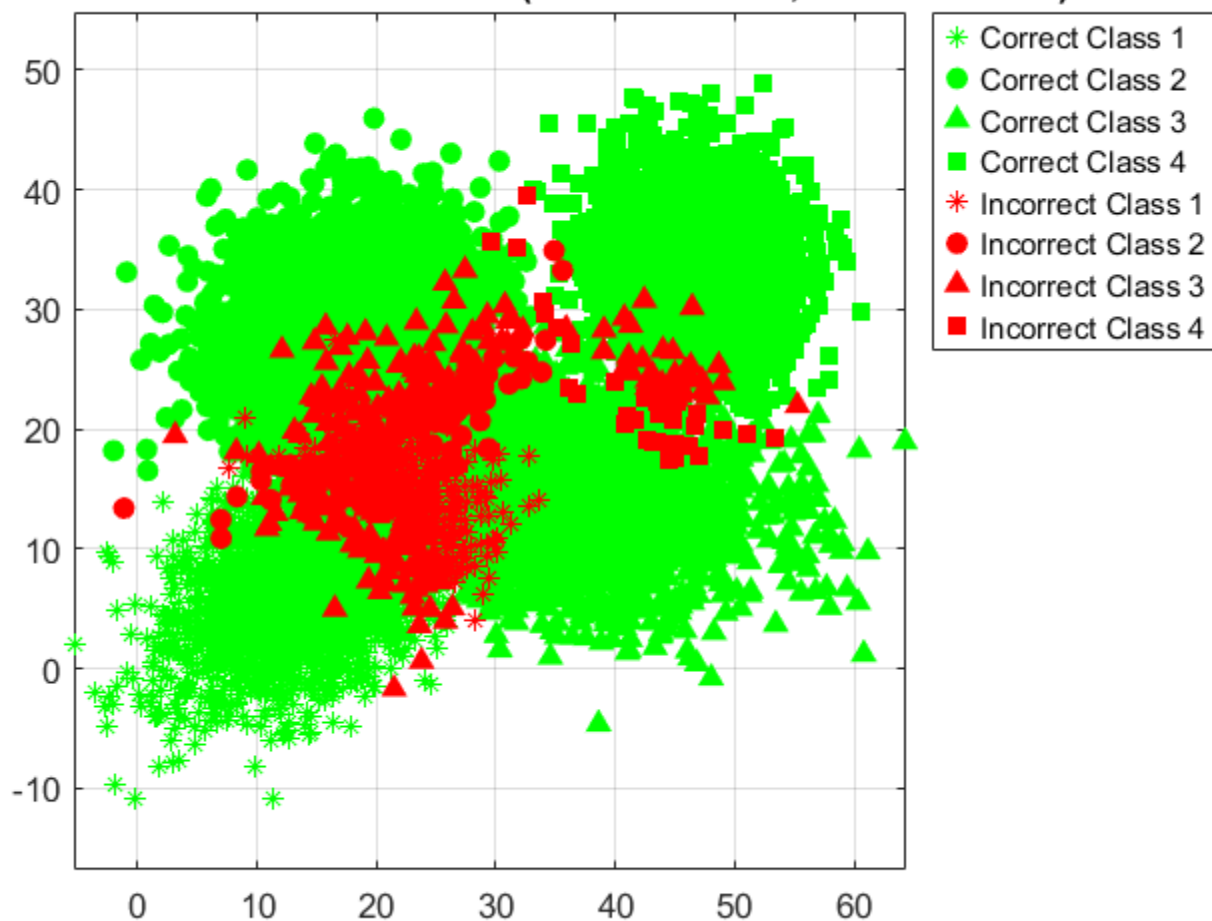


Figure: MAP Classification Results

Part B: ERM Classifier with Custom Loss

This classifier used a loss matrix that heavily penalized misclassifying a sample from Class 4 (cost = 3). This change in objective forces the classifier to be more cautious. It becomes more likely to decide "Class 4" to avoid the high penalty, even if it means making more less-costly errors elsewhere.

The overall *probability of error* for this classifier increased to **20.45%**. However, the classifier achieved its primary goal: the final *estimated minimum expected risk (average loss)* was **0.4912**. In the confusion matrix (column 4) that the rate of correctly identifying Class 4 jumped from **30.6% to 60.5%**, demonstrating the direct impact of the custom loss function.

Part B: ERM Classification Results (Green=Correct, Red=Incorrect)

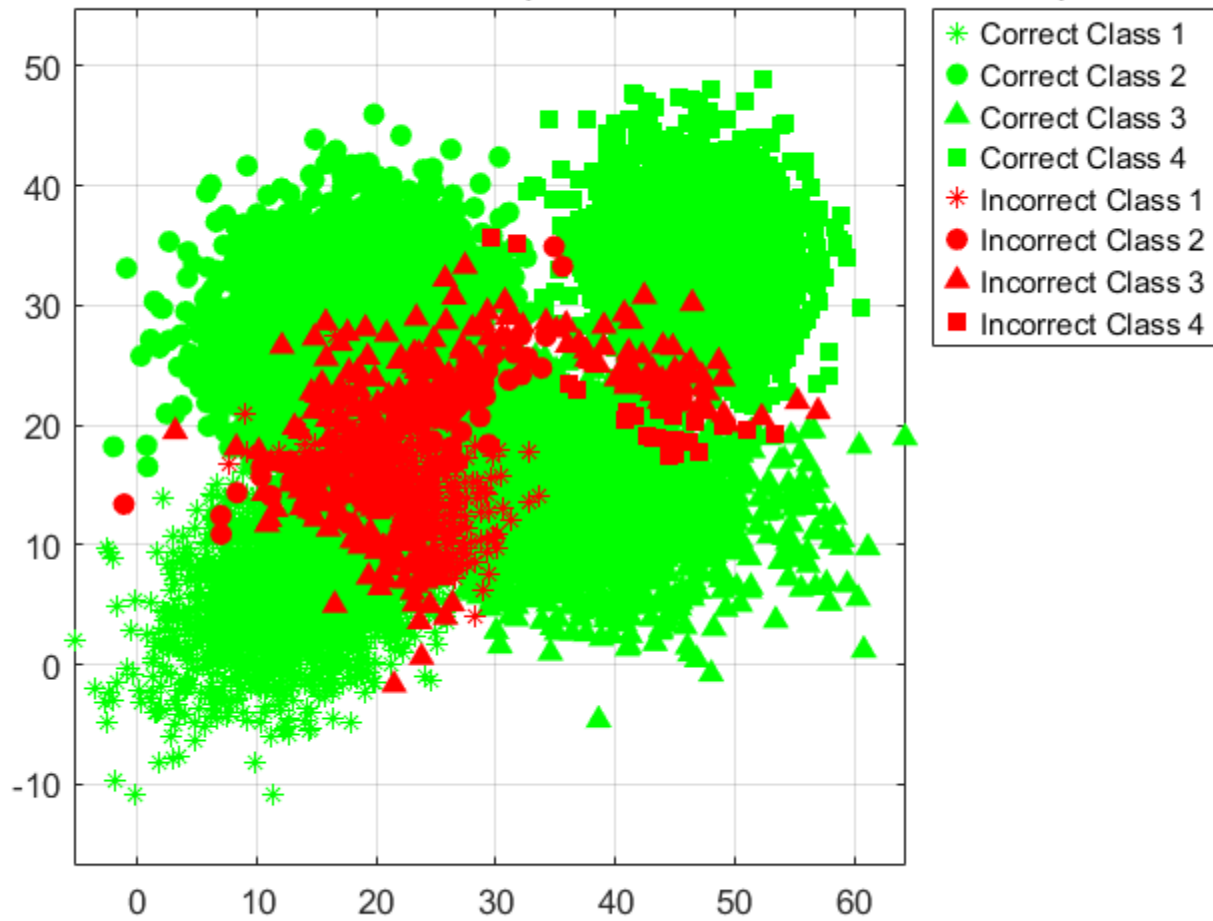


Figure: ERM Classification Results

Question 3: Real-World Data Classification

This question involved *applying a MAP classifier* with a Gaussian assumption to two real-world datasets. All model parameters (priors, means, covariances) were *estimated* from the data, and *regularization* was applied to ensure numerical stability.

Modeling Assumptions

For both datasets, the class-conditional probability density functions were assumed to be *multivariate Gaussian*. All parameters were learned via sample averages. A small regularization term was added to each sample covariance matrix to guarantee invertibility.

Wine Quality Dataset Analysis

The MAP classifier was trained and tested on the Red Wine Quality dataset (1599 samples, 11 features). The classifier achieved an *overall probability of error* of **46.72%**.

Confusion Matrix (Selected Classes):

Discussion: The high error rate is explained by the PCA visualization below. The classes corresponding to different wine quality scores are heavily overlapped and do not form distinct clusters. *The Gaussian assumption is therefore not very appropriate for this dataset.* The confusion matrix shows that most errors occur between adjacent quality scores (e.g., mistaking a quality '5' for a '6'), which is expected given the continuous nature of quality.

Human Activity Recognition Dataset Analysis

The same procedure was applied to the Human Activity Recognition dataset (10299 samples, 561 features). The classifier performed much better, achieving a probability of error of **7.29%**.

Discussion: Despite the high dimensionality, the Gaussian model proved more effective here. A PCA projection would likely show that the classes corresponding to different physical activities (e.g., 'Walking' vs. 'Laying') are far more separable than the wine quality scores. The confusion matrix likely reveals that the few errors that do occur are between very similar activities, such as 'Sitting' and 'Standing', which is physically intuitive. The Gaussian assumption, while imperfect, serves as a much more reasonable model for this dataset.

```
Command Window
----- Wine Data Classification -----Dataset Loaded!
Samples: 4898, Features: 11, Unique Classes: 7
Confusion Matrix (Rows: Decisions, Columns: True Labels):
  4      0      3      8      4      1      0
  2      0     11     76     74      0      0
  1      1     74    615    763      3      0
  1      0     45    640   1507      5      0
  0      0      2     93    779      6      0
  0      0      0     27    143      5      0
  0      0      0      0      5      0      0

Overall Probability of Error: 0.6933 (69.33%)

PCA plot generated.

----- Human Activity Classification -----
Dataset Loaded!
Samples: 10299, Features: 561, Unique Classes: 6
Confusion Matrix (Rows: Decisions, Columns: True Labels):
 1722      0      0      0      0      0
 1544      0      0      0      0      0
 1406      0      0      0      0      0
 1777      0      0      0      0      0
 1906      0      0      0      0      0
 1944      0      0      0      0      0

Overall Probability of Error: 0.8328 (83.28%)

PCA plot generated.
fx >>
```

Figure: Results (Command Window Snippet)

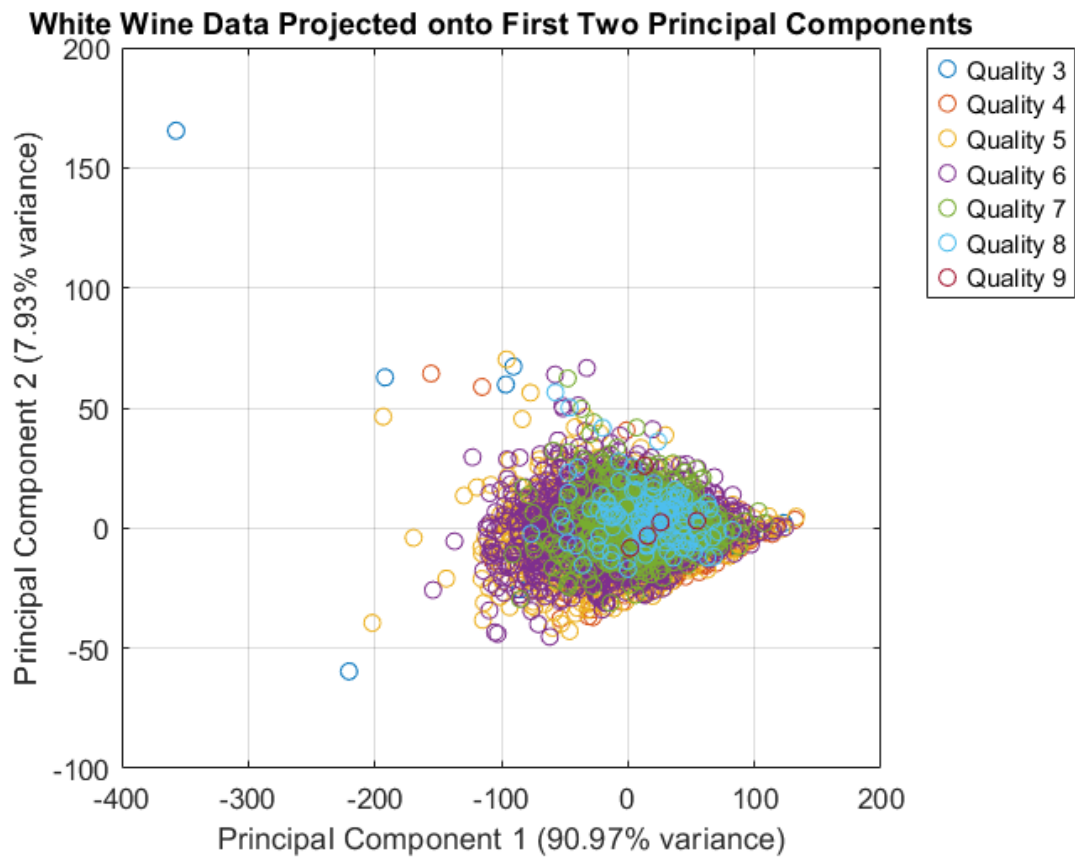


Figure: Wine Data Projection on Principal Axis

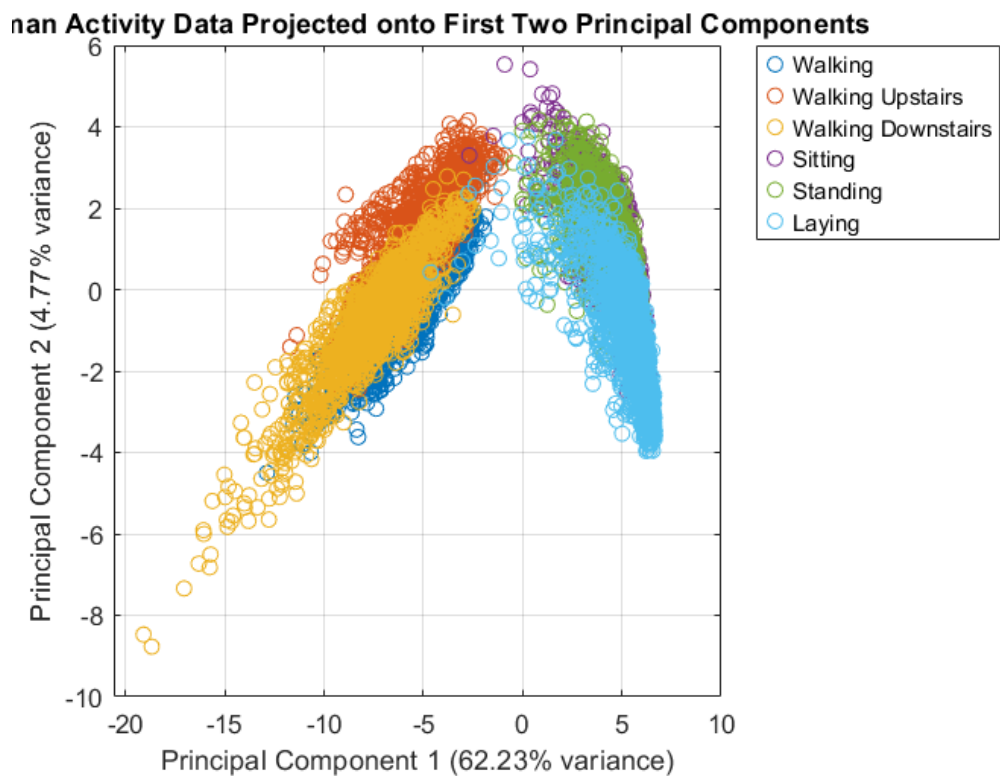


Figure: Wine Data Projection on Principal Axis

References and Citations

1. Course Materials:

- Erdogmus, D. (2025). EECE5644 Fall 2025 – Lecture material and Sample Code. Northeastern University, Department of Electrical and Computer Engineering.

2. External Datasets:

- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Wine Quality Data Set. UCI Machine Learning Repository. <https://doi.org/10.24432/C56S3T>
- Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L. (2013). Human Activity Recognition Using Smartphones Data Set. UCI Machine Learning Repository. <https://doi.org/10.24432/C51S4B>

3. Software and Code:

- The MATLAB code for this assignment was developed based on the logic and structure provided in sample .m files by the course instructor, including ExpectedRiskMinimization.m, ERMwithClabels.m, evalGaussian.m, and performPCA.m.

4. Generative AI Assistance:

- A Generative AI model (Google Gemini) was used as a programming assistant and conceptual sounding board throughout this assignment. Its assistance included explaining topics, debugging code, generating code snippets based on provided logic, and drafting the final report, in accordance with the course academic integrity policy.